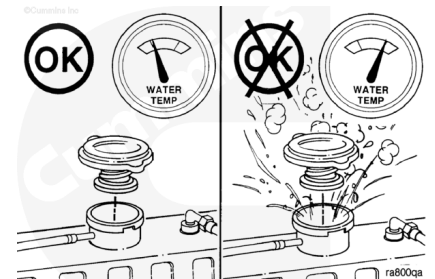


Trainings ▾

Drain

⚠ WARNING ⚠

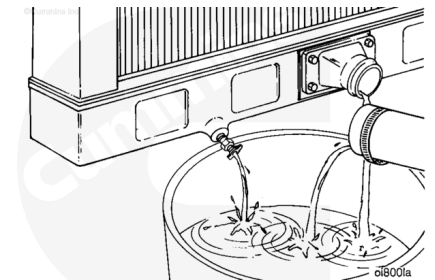
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Remove the radiator cap after the engine is cool.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

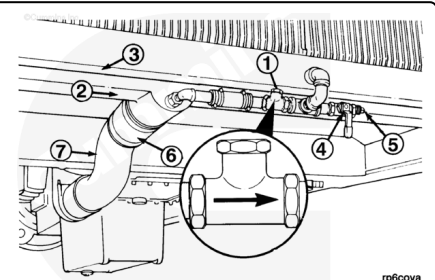


Position the vehicle on level ground. Open the draincock at the bottom of the radiator. Remove the lower radiator hose. Drain the cooling system.

LTA

⚠ WARNING ⚠

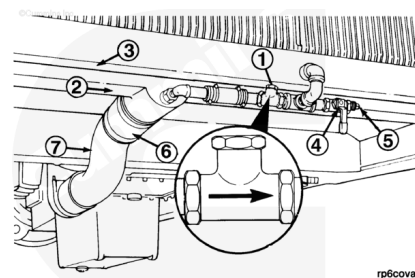
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Position the vehicle on level ground. Drain the coolant.

The low temperature aftercooler (LTA) cooling system is drained by removing a pipe plug (5) and opening a swing valve (4) located in the coolant plumbing connecting the bottom tank of the main engine and LTA radiator cores.

Note : On some radiators, the plumbing that connects the bottom tank of the engine and the low temperature aftercooler (LTA) radiator is incorporated internally. With this design, there will be two drain locations on the radiator bottom tank.



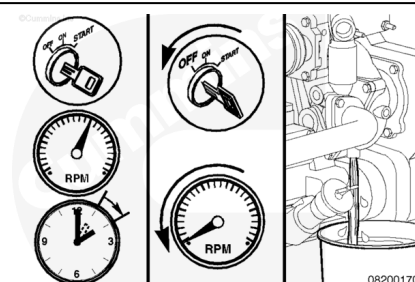
The plumbing between the radiator cores is designed to **only** allow coolant to flow from the main engine core (2) to the LTA core (3) during the fill procedures. To prevent coolant flow in the opposite direction during engine operation, a swing valve (1) **must** be included in the plumbing. The arrow on the swing valve **must** point toward the LTA core.

Draining the cooling system at the main engine radiator core by removing the hose (6) or the pipe (7) will **not** allow the LTA core to drain due to the swing valve.

Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

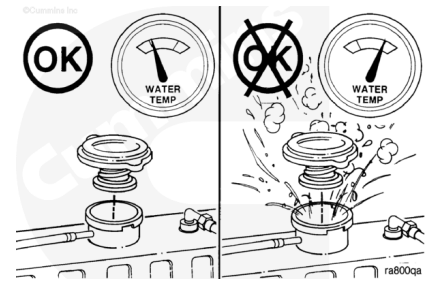
Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1 ½ hours.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

Flush

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

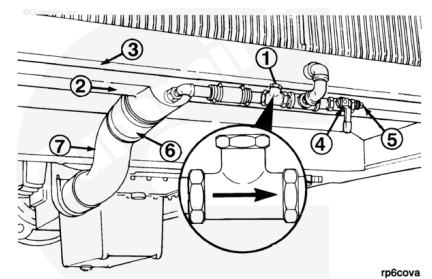


Note : The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning time. Restore™ can be safely used up to twice the recommended concentration levels. Extremely scaled or fouled systems can require more than one cleaning.



Drain the cooling system.

Do **not** remove the coolant filters.

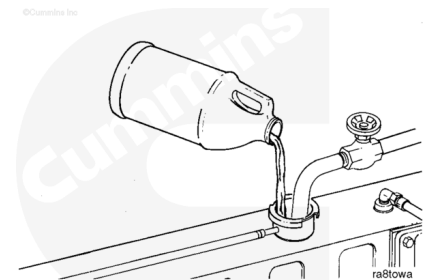


⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

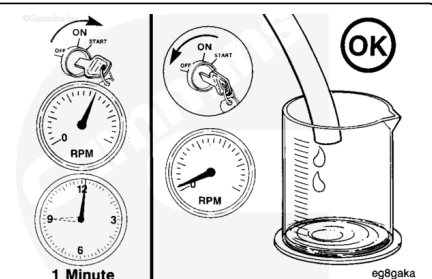
Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, or equivalent, for each 38 to 57 liters [10 gal] of cooling system capacity, and fill the system with plain water.

Turn the heater temperature switch to “HIGH” to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

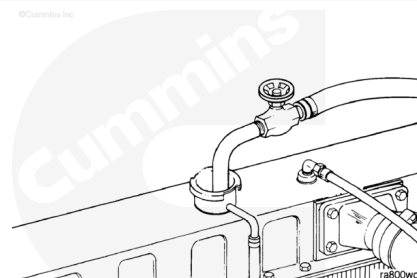


Operate the engine at a coolant temperature of at least 85°C [185°F] for 1 to 1-½ hours.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.



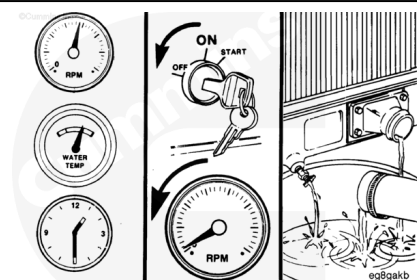
Fill the cooling system with clean water.



Operate the engine at high idle for 5 minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

If the water drained is still dirty, the system **must** be flushed again until the water is clean.

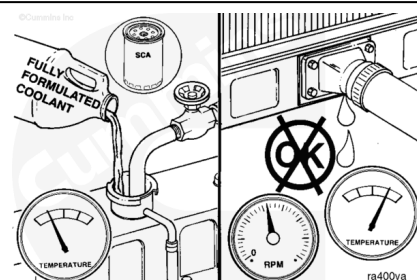


Install a new coolant filter and fill the cooling system with fully formulated coolant. Refer to Procedure 008-006 in Section 8.

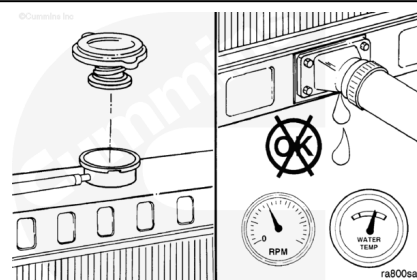
(/qs3/pubsys2/xml/en/procedures/20/20-008-006-tr.html)

Refer to Procedure 008-060 in Section 8.

(/qs3/pubsys2/xml/en/procedures/20/20-008-060.html)



Install the pressure cap. Operate the engine until the coolant reaches a temperature of 70°C [160°F], and check for coolant leaks.



Pressure Test

Check the coolant level and fill if necessary. Refer to Procedure 008-018 in Section 8

(/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)

To aid in determining the location of coolant leak, it could be necessary to add fluorescent tracer, Part Number 3377438, to the coolant.

Add one unit of fluorescent tracer to each 38 liters [10 gal] of coolant.

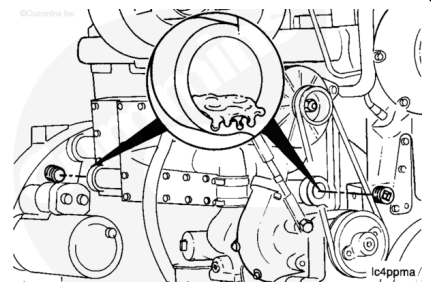
Idle the engine for 5 to 10 minutes or until normal operating temperature is reached, to allow the dye to circulate through the cooling system.

Use a high intensity black light, Part Number 3163337, or equivalent, to illuminate the dye.

If the customer experiences a coolant loss that can **not** be attributed to an external coolant leak or to coolant being pushed out of the overflow.

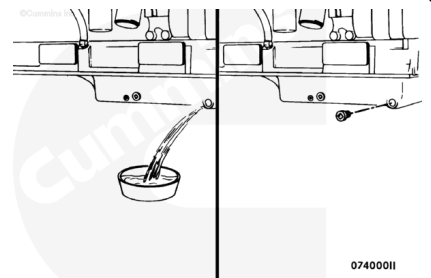
Internal

Check the oil cooler housing for internal coolant leakage. Remove the plugs from the cooler housing and check for coolant. If coolant is present, replace the oil cooler elements. Pressure test the elements to determine cause of leakage. Refer to Procedure 007-007 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-007-tr.html)



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



If the oil cooler cores are **not** leaking, drain the oil. Refer to Procedure 007-037 in Section 7.

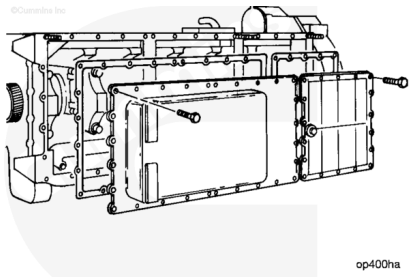
(/qs3/pubsys2/xml/en/procedures/20/20-007-037.html)

Check for coolant in the oil.

If the oil does **not** contain coolant, check the aftercooler. Refer to Procedure 010-002 in Section 10.

(/qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html)

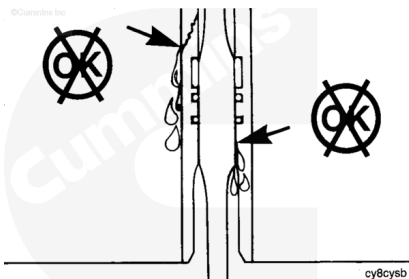
If the oil contains coolant, remove the oil pan and adapter cover. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-037.html) Refer to Procedure 007-026 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html)



Check for leakage on the inside and outside of the cylinder liners.

Replace any cylinder liners that are leaking. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html)

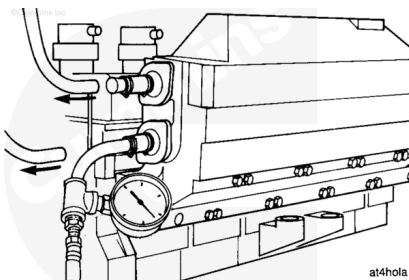
If the cylinder liners are **not** leaking, check the aftercooler element.



Disconnect the aftercooler coolant hoses. Plug one hose and apply air pressure to the other hose.

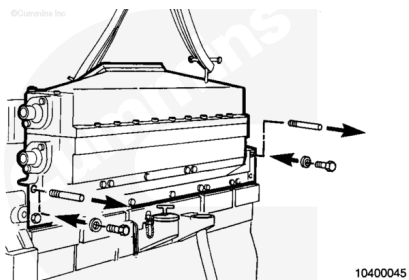
Measurements		
	kpa	psi
Air Pressure	415	60

If the air pressure decreases, replace the aftercooler element. Refer to Procedure 010-008 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-008.html)



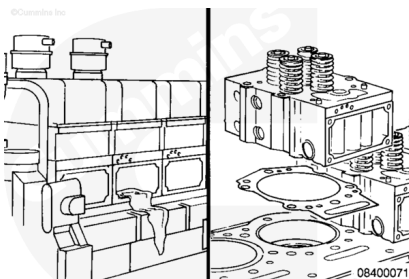
If the aftercooler element is **not** leaking, check the cylinder heads.

Remove the aftercooler housing. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html)



Check the intake ports and exhaust port for coolant/fluorescent dye using the black light. The presence of fluorescent dye indicates a crack.

Note : A green glow would be seen in the area where the leak has occurred. It is the coolant with tracer dye glowing from the light of the black light.



Remove the valve covers. Shine the black light in the overhead around the valve springs. If a spring deck crack is present and leaking coolant, it will glow under the black light.

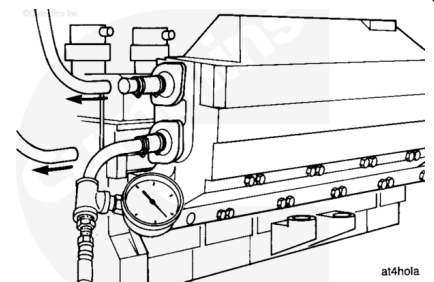
Not all cracks will easily be detected by this procedure. It is also possible to detect signs of coolant in the exhaust by looking for sign of either rust or evidence of steam cleaning in either the exhaust port in the head or its corresponding position in the exhaust manifold.

If coolant or fluorescent dye is found in the intake/exhaust ports, remove the cylinder head and test for defects. See the following procedure to replace the cylinder head. Refer to Procedure 002-004 in Section 2.

(/qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html)

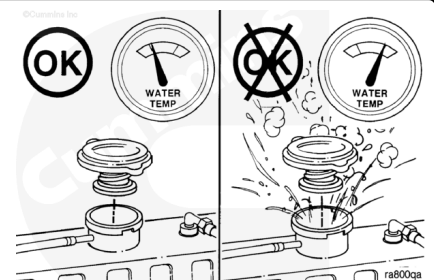
Remove the test equipment. Remove the plug from the overflow tube.

Replace the pressure cap.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

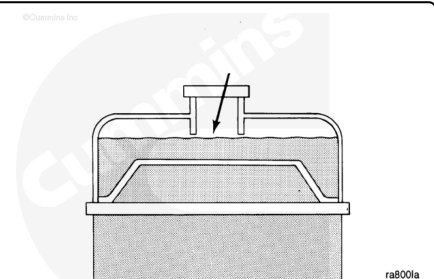


To confirm and locate external coolant leaks, it is **not** necessary to warm the engine.

External

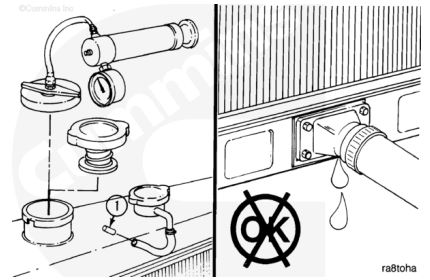
Check the coolant level and fill if necessary.

To confirm and locate external coolant leaks that occur during normal engine operation, complete this test while the engine is warm.



⚠ CAUTION ⚠

Do not apply more than 140 kPa [20 psi] air pressure to the cooling system. Excessive air pressure will damage the water pump seal.



If the radiator is equipped with a pressure relief valve, install a plug in the overflow tube (1).

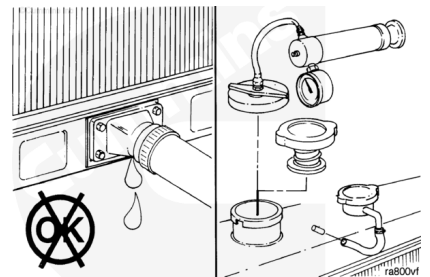
Install the pressure tester on the radiator fill neck or surge tank, if equipped. Apply maximum air pressure.

Measurements

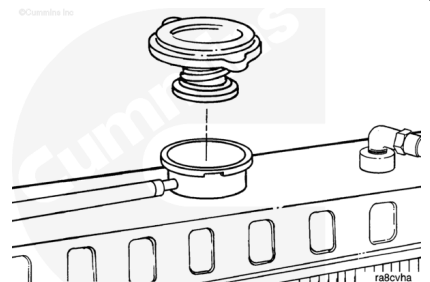
	kpa	psi
Maximum Air Pressure	140	20

Maintain the air pressure for a minimum of 5 minutes. Check for leaks. Any leaks **must** be repaired.

Remove the pressure test equipment and the plug from the pressure relief valve overflow tube, if equipped.



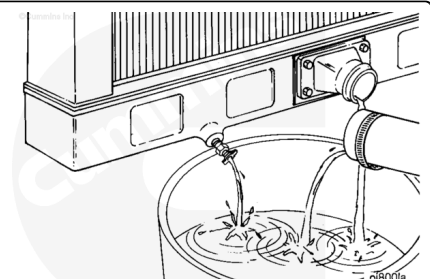
Replace the coolant system pressure cap.



Fill

⚠ CAUTION ⚠

Use caution when filling the cooling system so that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations.



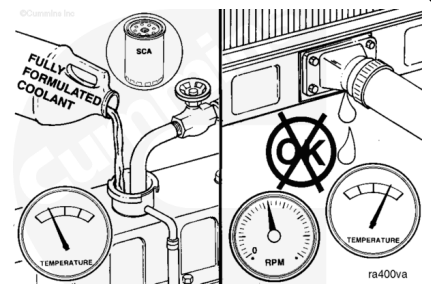
Close the radiator draincocks.

Install the lower radiator hose(s). Tighten the hose clamps.

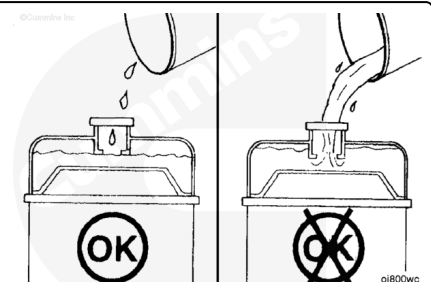
Torque Value: 5 n•m [40 in-lb]

Use fully formulated coolant to fill the cooling system.

Use the correct units of SCA to obtain the correct cooling system protection. Refer to Procedure 008-060 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-060.html>)

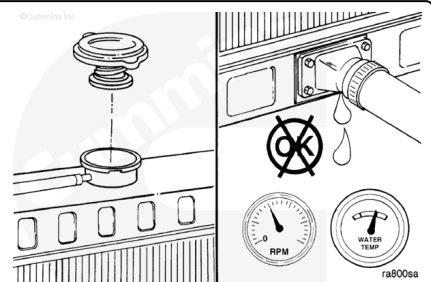


Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.



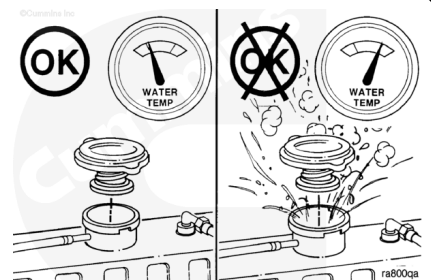
Replace the radiator or fill cap.

Operate the engine until the coolant reaches a temperature of 70°C [160°F]. Check for leaks.



⚠ WARNING ⚠

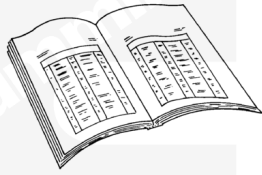
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Shut the engine OFF and allow it to cool.

Rail Applications

Refer to original equipment manufacturer (OEM) service manual.



ck800wa

Last Modified: 07-Apr-2015
